# Release Bundle

FirmwareSight 0.6.0 · release `release-e400a8ac51a57d34e309d254536f8850044ef4dd53e26245045c24fdc86fdcc5`

## 1. Release identity

Four different identities, named separately on purpose.

| | |
|---|---|
| **Release id** | `release-e400a8ac51a57d34e309d254536f8850044ef4dd53e26245045c24fdc86fdcc5` |
| **Project release version** | `1.2.3` — resolved from `release.expected_version` |
| **FirmwareSight app version** | `0.6.0` |
| **Snapshot analyzed** | `snap-4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374-p0-normalize-1-a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6` |
| **Analysis normalization rule** | `p0-normalize-1` |
| **Workspace HEAD observed by the Release Gate** | `585dafd3eb59592563923818cdfec4e9a370a83b` |
| **Workspace tag observed by the Release Gate** | `v1.2.3` |
| **Workspace dirty observed by the Release Gate** | false |

These Git facts describe the workspace the Release Gate ran in. They are not evidence about how the shipped bytes were compiled, and no claim of that kind is made here.

## Release visual overview




Visual summary only. Detailed evidence follows. Browser Print / Save as PDF; native PDF export is unavailable.

## 2. Shipped artifacts

2 file(s) under `artifacts/`. The baseline build is never shipped: it exists only to explain the comparison in section 4.
The bytes this release consists of, in the order the release fingerprint lists them.

| Bundle path | Kind | SHA-256 | Bytes |
|---|---|---|---|
| `artifacts/firmware.elf` | elf | `4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374` | 17464 |
| `artifacts/firmware.map` | map | `a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6` | 4664 |

## 3. Analysis summary

What was read off the build, per input file.

| File | Format | Entry point | Build id |
|---|---|---|---|
| `firmware.elf` | Arm 32-bit little · `elf` | `0x8000049` | no .note.gnu.build-id section is present |

| File | Format | Entry point | Build id |
|---|---|---|---|
| `firmware.map` | Unknown unknown-bit unknown · `map` | a linker MAP records addresses, not an entry point | a linker MAP carries no build-id |

### Memory budgets

Absence is printed as absence. A budget that is not a number was not measured, and is never shown as zero.

| | |
|---|---|
| **Nonvolatile / load image** | 376 bytes (exact) |
| **Runtime RAM** | 76 bytes (exact) |
| **Accounting rule** | `adr-0021-dual-budget` |
| **Layout source** | `map` |
| **Weakest evidence basis** | map-memory-configuration+elf-load |
| **Admissible for a hard block** | true |
| **Device metadata held out of both budgets** | 4294 bytes |

### Coverage of this build

| | |
|---|---|
| **Rows carried in the analysis document** | 19 section(s), 52 symbol(s), 11 evidence record(s) |
| **ELF support** | `supported` |
| **Sections** | `available` |
| **Symbols** | `available` |
| **Debug info** | `available` |
| **MAP companion** | `provided` |
| **Object attribution** | `unavailable` |
| **Git facts** | `unknown` |

## 4. Compare summary

Totals moved between the two builds. The full row-by-row comparison is in `diff.json`.

| | |
|---|---|
| **Base (old) snapshot** | `snap-3f615b6247179d930f59b76dcdbaea1a5eca8c835ab6d410feb20a062f33f21f-p0-normalize-1-832690060a8d25f8acacd85a62ce76bc83dfb6768e435bcc04c51f9664ed363e` |
| **Target (new) snapshot** | `snap-4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374-p0-normalize-1-a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6` |
| **Nonvolatile / load image** | 256 bytes → 376 bytes · delta +120 (exact) |
| **Runtime RAM** | 8 bytes → 76 bytes · delta +68 (exact) |
| **Comparability of the pair** | exact |
| **Rows that differ** | sections: 1 added, 1 removed, 16 changed, 0 unpaired · symbols: 38 added, 33 removed, 5 changed, 67 unpaired |
| **Identical, so left out of the tables** | 2 section(s), 9 symbol(s) |

### Top growth, as Core ranked it

Top section growth

| Delta | Row | Bytes | Change |
|---:|---|---:|---|
| +221 | `.debug_loclists` | 414 | Changed |
| +112 | `.debug_info` | 910 | Changed |
| +107 | `.debug_line` | 421 | Changed |
| +88 | `.text` | 212 | Changed |
| +80 | `.symtab` | 848 | Changed |

Top symbol growth

| Delta | Row | Bytes | Change |
|---:|---|---:|---|
| +42 | `mqtt_task` | 46 | Changed |
| +16 | `payload_extra` | 20 | Changed |

The complete comparison ships as `diff.json` under schema `urn:firmwaresight:schema:diff:1` version 1.

## 5. Release Gate

✓ PASS — the disposition that made this bundle possible. A bundle is written only for a **PASS** after accepted reviews; a **REVIEW** or a **BLOCK** produces no release, and an evidence gap never becomes a review.

| | |
|---|---|
| **Run id** | `gate-a17e9861ce643da5e7061ac349793f80d1be87736ad52b2daf3433dc8cb8bc26` |
| **Snapshot** | `snap-4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374-p0-normalize-1-a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6` |
| **Baseline** | `snap-3f615b6247179d930f59b76dcdbaea1a5eca8c835ab6d410feb20a062f33f21f-p0-normalize-1-832690060a8d25f8acacd85a62ce76bc83dfb6768e435bcc04c51f9664ed363e` |
| **Policy digest** | `b87aa057854d8a0089594ad9445b0626174dc0308d636dd6d6854902ae1e5e98` |
| **Project config schema** | `1` |
| **Severity before reviews** | ✓ PASS |
| **Severity after accepted reviews** | ✓ PASS |

### Findings by state

Core's counts for this run, carried rather than recounted.

| State | Count |
|---|---|
| ✕ BLOCK | 0 |
| ▲ REVIEW | 0 |
| ? UNKNOWN | 0 |
| ✓ PASS | 8 |
| – N/A | 2 |

### All findings

In the order a release owner reads them: what blocks, what needs a decision, what is unknown, what passed. Every finding of the run is listed, not a truncated top list.

| Rule | State | Effective severity | Finding | Evidence | Next step |
| --- | --- | --- | --- | --- | --- |
| `git.clean` | ✓ PASS | same as state | Workspace has no uncommitted changes at Gate evaluation time. | `git:status` · `git:head` | none recorded |
| `release.version_matches_policy` | ✓ PASS | same as state | Workspace tag `v1.2.3` matches the version pattern and its version `1.2.3` is the declared release version. | `policy:version.source` · `policy:version.pattern` · `git:exact-tag=v1.2.3` · `policy:release.expected_version` | none recorded |
| `artifacts.required` | ✓ PASS | same as state | Every required artifact kind is present in the snapshot (elf, map). | `policy:artifacts.required=elf,map` · `artifact:elf:4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374` · `artifact:map:a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6` | none recorded |

| Rule | State | Effective severity | Finding | Evidence | Next step |
| --- | --- | --- | --- | --- | --- |
| artifacts.hashes | ✓ PASS | same as state | All 2 snapshot artifact(s) carry a SHA-256 digest. | artifact:elf:4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374 · artifact:map:a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6 · policy:artifacts.hashes | none recorded |
| memory.flash_budget | ✓ PASS | same as state | Nonvolatile footprint 376 B is within the 512 B budget (136 B headroom). | policy:memory.flash_budget=512 · evidence:ev-memory-nonvolatile_image_footprint_bytes | none recorded |
| memory.ram_budget | ✓ PASS | same as state | Runtime RAM footprint 76 B is within the 256 B budget (180 B headroom). | policy:memory.ram_budget=256 · evidence:ev-memory-runtime_ram_footprint_bytes | none recorded |

| Rule | State | Effective severity | Finding | Evidence | Next step |
| --- | --- | --- | --- | --- | --- |
| diff.growth | ✓ PASS | same as state | Growth against the baseline stays within every configured threshold. | policy:diff.flash_growth_review_bytes=200 · diff:snap-3f615b6247179d930f59b76dcdbaea1a5eca8c835ab6d410feb20a062f33f21f-p0-normalize-1-832690060a8d25f8acacd85a62ce76bc83dfb6768e435bcc04c51f9664ed363e:snap-4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374-p0-normalize-1-a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6:nonvolatile | none recorded |

| Rule | State | Effective severity | Finding | Evidence | Next step |
|---|---|---|---|---|---|
| release.notes | ✓ PASS | same as state | Release Notes `docs/RELEASE_NOTES.md` is present. | policy:release.require_release_notes · file:docs/RELEASE_NOTES.md · file:docs/RELEASE_NOTES.md#sha256=d547160abce41f59ec545b2013c8f0491f1fe7061834e374e1cdf376ed78f93d | none recorded |
| release.commit_matches_expected | – N/A | ✓ PASS | No `release.expected_commit` is declared for this release. | policy:release.expected_commit | none recorded |
| evidence.unknown_review | – N/A | ✓ PASS | `gate.unknown_evidence_review_count` is not configured. | policy:gate.unknown_evidence_review_count | none recorded |

## 6. Accepted reviews

No review was accepted for this run, so the disposition in section 5 is the one Core computed with nothing set aside. An empty list is a fact, not a missing export.

## 7. Release notes

| | |
|---|---|
| **Included** | yes, as release-notes.md |
| **SHA-256 of the shipped copy** | d547160abce41f59ec545b2013c8f0491f1fe7061834e374e1cdf376ed78f93d |
| **Bytes** | 151 |

The notes ship as their own file and are not reproduced inside this report. Markdown a reader wrote is never parsed, rewritten or embedded as raw HTML by the bundle, which is what keeps this document free of markup a note could carry.

## 8. Bundle integrity

Two layers of SHA-256 lists cover this bundle. `SHA256SUMS` hashes every payload file except itself and the manifest. `release-manifest.json` hashes every file except itself, that list included.
Every file a digest can be named for here. This report is payload: it is written before both index files, so it cannot print their digests, or its own.

| Bundle path | SHA-256 | Bytes |
|---|---|---|
| `accepted-reviews.json` | `50afd7fab31bbd69f5fb91a5357f3128544e76696b5e3562081acd58e23e0076` | 170 |
| `analysis.json` | `ba20d979db4c92fd98780f8afc5186e0e51d7a8dfef8f40eb8ad6c3786bf199a` | 26963 |
| `artifacts/firmware.elf` | `4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374` | 17464 |
| `artifacts/firmware.map` | `a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6` | 4664 |
| `diff.json` | `1930cdb6a098c67456297ec321910315d21aeb3ce79ef7264fcc4037e148242a` | 64355 |
| `gate-results.json` | `278a1a63c5cd81f86319f9393762a17de8ad69d374fa52649ab403738d86ba23` | 4803 |
| `release-notes.md` | `d547160abce41f59ec545b2013c8f0491f1fe7061834e374e1cdf376ed78f93d` | 151 |
| `release-report.html` | indexed by `release-manifest.json`, which is written after this report | |
| `SHA256SUMS` | indexed by `release-manifest.json`, which is written after this report | |
| `release-manifest.json` | indexed by `release-manifest.json`, which is written after this report | |

### The self-reference rule

| | |
|---|---|
| **`SHA256SUMS` covers** | every bundle file except itself and the manifest |
| **`SHA256SUMS` excludes** | `SHA256SUMS`, `release-manifest.json` |
| **`release-manifest.json` covers** | every bundle file except itself, SHA256SUMS included |
| **`release-manifest.json` excludes** | `release-manifest.json` |

A file cannot contain its own cryptographic hash without a fixed-point scheme, so no self-digest is faked: `self_digest_written` is false. no file carries its own digest; SHA256SUMS hashes the payload, release-manifest.json hashes the payload plus SHA256SUMS

**Verifying this bundle without FirmwareSight**

Every file here is a plain file. Any SHA-256 tool and any text editor is enough. The check below proves that the bytes match the lists — an integrity and consistency statement about the bundle as written. It says nothing about who wrote it, and nothing in this bundle is a signature.

1. List the directory. It must contain exactly the files `release-manifest.json` indexes, plus that manifest itself, and no temporary or hidden file.
2. Recompute the SHA-256 of every file named in `SHA256SUMS` and compare it with the digest on that line. On a POSIX host one command does the whole list: `sha256sum -c SHA256SUMS`.
3. On Windows, `Get-FileHash -Algorithm SHA256 <path>` prints the same digest in upper case; lower-case it before comparing, because both lists are written in lowercase hex.
4. Compare every line of `SHA256SUMS` with the entry of the same path in the `files[]` array of `release-manifest.json`. Digests and sizes must agree, and the manifest must additionally carry an entry for `SHA256SUMS` itself.
5. Confirm that neither list contains a blank digest, an all-zero digest, or the digest of the file it sits in. There should be none of those, and looking for them is what makes the rule in the table above checkable rather than a claim.
6. Open `analysis.json`, `gate-results.json` and `accepted-reviews.json` as plain JSON. Each names the schema it is an instance of, and the manifest's `extensions` block states which schema majors this bundle was written against.

## 9. Known capability limits of this release

What the inputs of this build did not allow FirmwareSight to establish. A limit is stated as a limit instead of being left as an empty cell.
Capability reporting, taken from the analysis document in this bundle.

| Capability | Reported | What that means for this release |
|---|---|---|
| Object attribution | `unavailable` | Totals are reported per section and per symbol only. No per-object-file or per-module total is claimed here, and none should be inferred from the section rows. |
| Git facts | `unknown` | This capability is reported at less than full coverage by the analysis document in this bundle. |

**Object and module attribution**

the persisted symbol and section records carry no object-file or module attribution, so no object or module delta is claimable for either build

**Comparison warnings**

- `SYMBOL-AMBIGUOUS` — 67 symbol row(s) repeat a name, kind and binding combination on one or both sides and were left unpaired; a row position is not an identity

Deterministic export: no timestamp, no scripts, no network requests. Written by FirmwareSight 0.6.0 over the documents in this bundle.